**Mapping (Version 1.0)**

```
Crypto_Verify(publicKey, message, signature) :=
     boolean ECDSAVerify(QU := publicKey, M := message, S := signature)
```

`ECDSAVerify()` SHALL be the ECDSA signature verification function as defined in Section 4.1.4 of SEC 1 using `Crypto_Hash()` as the underlying hash function `Hash()`; returns `TRUE` if the verification succeeds and `FALSE` otherwise.

### 3.5.4. ECDH

`Crypto_ECDH()` is used to compute a shared secret from the Elliptic Curve Diffie-Hellman (ECDH) protocol.

**Function and description**

```
byte[CRYPTO_GROUP_SIZE_BYTES]
Crypto_ECDH(
     PrivateKey myPrivateKey,
     PublicKey theirPublicKey)
```

Computes a shared secret using Elliptic Curve Diffie-Hellman.

**Mapping (Version 1.0)**

```
Crypto_ECDH(myPrivateKey, theirPublicKey) :=
     byte[CRYPTO_GROUP_SIZE_BYTES] ECDH(dU := myPrivateKey, QV := theirPublicKey)
```

The output of `ECDH()` SHALL be the serialization of the x-coordinate of the resultant point as defined in Section 3.3.1 of SEC 1.

### 3.5.5. Certificate validation

`Crypto_VerifyChain()` is used to verify Matter certificates.

`Crypto_VerifyChainDER()` is used to verify public key X.509 v3 certificates in X.509 v3 DER format.

**Function and description**

```
boolean
Crypto_VerifyChain(MatterCertificate[] certificates)
```

Given Matter certificates, `Crypto_VerifyChain()` performs all the necessary validity checks on `certificates`, taking in account that the notion of "current time" for the purposes of validation SHALL abide by the rules in Section 3.5.6, “Time and date considerations for certificate path validation”.

 